

Machine Learning A (2026–2027)

Home Assignment 3

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1 Optimal Classification (16 points)

1.1 Question 1

1.2 Question 2

1.3 Question 3

2 Logistic Regression (50 points)

2.1 Cross-entropy error measure (16 points)

2.1.1 Rewriting negative logarithmic likelihood as cross-entropy

2.1.2 Cross-entropy and logistic regression

2.2 Logistic regression loss gradient (24 points)

2.2.1 Gradient for -1 and 1

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2.2.3 Influence of misclassified examples

2.3 Log-odds (10 points)

3 Sleep Well (34 points)

3.1 Data understanding and preprocessing (4 points)

3.2 Classification (30 points)

3.2.1 Logistic regression

3.2.2 Random forest

3.2.3 Nearest neighbor

4 Overfitting (optional, 0 points)